



Telemetry-Aware Model Monitoring & Prompt Optimization

this is the topic for my final yr disertation and final yr project

of which as of now i have to submit the following:

tell me about the project domain

Introduction focussing on what our domain is an how our projects corelates to the domain

Literature Survey

(Focus on: Why do you select the specific domain?)

(Atleast 30 International Journal/Conference papers should be mentioned. 80% of the papers should be from last 5 years)

Gap Identification from the paper survey

(Focus on: How do you identify the problem statement of your project?)

Based on the comprehensive research gathered, I can provide detailed information about your dissertation topic "Telemetry-Aware Model Monitoring & Prompt Optimization" along with the required components.

Project Domain

The domain encompasses **AI Operations (AIOps)** and **Machine Learning Operations (MLOps)** with a specific focus on intelligent monitoring systems for AI/ML applications. This interdisciplinary field combines traditional telemetry monitoring with modern AI model observability, creating systems that can autonomously track, analyze, and optimize machine learning models in production environments.^{[1] [2] [3] [4]}

Introduction

The project domain centers on **telemetry-aware intelligent systems** that monitor and optimize AI model performance in real-time production environments. As organizations increasingly deploy large language models (LLMs) and complex ML systems at scale, there's a critical need for sophisticated monitoring infrastructure that goes beyond traditional system metrics to understand model behavior, prompt effectiveness, and user interaction patterns.^{[5] [6] [7] [1]}

Traditional monitoring approaches focus on system-level metrics like CPU usage and response times, but fail to capture the nuanced performance characteristics of AI models. This project domain addresses the gap by integrating **telemetry data collection** with **AI-driven analysis capabilities**, enabling systems to detect model drift, optimize prompts automatically, and maintain model performance without human intervention.^{[2] [3] [8] [1]}

The correlation between telemetry-aware monitoring and prompt optimization represents a paradigm shift toward **autonomous AI system management**. By leveraging real-time telemetry streams, these systems can identify when prompts are underperforming, automatically generate improved versions, and deploy optimizations through continuous integration pipelines.^[9] ^[10] ^[1]

Literature Survey Focus

Why This Domain Was Selected

The selection of telemetry-aware model monitoring and prompt optimization is driven by several critical industry needs and research gaps :^[11] ^[12]

Scalability Challenges: As organizations deploy hundreds of ML models in production, manual monitoring becomes impractical, necessitating automated, telemetry-driven solutions.^[8] ^[2]

Model Reliability: Studies show that ML models degrade over time due to data drift and changing environments, requiring continuous monitoring and optimization.^[13] ^[3]

Prompt Engineering Bottlenecks: Manual prompt optimization is time-consuming and inconsistent, creating demand for automated systems that can optimize prompts based on performance telemetry.^[10] ^[9]

Integration Gaps: Existing monitoring tools lack integration between system telemetry and model-specific metrics, creating blind spots in AI system observability.^[4] ^[6]

Key Research Areas (30+ Papers Required)

Model Monitoring and Drift Detection (10-12 papers)

- **Recent Surveys:** "Monitoring Machine Learning Systems: A Multivocal Literature Review" (2025) provides comprehensive coverage of ML monitoring practices^[11]
- **Drift Detection:** "Monitoring machine learning models: a categorization of challenges" (2022) categorizes key monitoring challenges^[14]
- **Healthcare Applications:** "A systematic literature review on Machine Learning Model Monitoring in Healthcare" (2023) shows domain-specific monitoring needs^[12]
- **Production Practices:** "Model Monitoring in Practice: Lessons Learned and Open Problems" (2022) discusses real-world implementation challenges^[15]

Telemetry and Observability (8-10 papers)

- **OpenTelemetry Integration:** Recent papers on unified observability frameworks for ML systems^[16] ^[4]
- **AI-Driven Telemetry:** "The AI-Driven Future of Real-Time Telemetry Analytics in Computer Networks" (2024) explores ML-enhanced telemetry processing^[17]
- **LLM Observability:** "An Introduction to Observability for LLM-based applications" (2025) covers specialized monitoring for language models^[18]

Prompt Optimization (8-10 papers)

- **Multi-Step Optimization:** "PRompt Optimization in Multi-Step Tasks (PROMST)" (2024) addresses complex prompt optimization challenges^[19]
- **System-Level Optimization:** "SPRIG: Improving Large Language Model Performance by System Prompt Optimization" (2024) focuses on general prompt optimization^[10]
- **Human Feedback Integration:** "Prompt Optimization with Human Feedback" (2024) incorporates user feedback into optimization loops^[20]
- **Automated Frameworks:** "An Automatic Prompt Optimization Framework for Large Language Models" (2025) presents fully automated optimization systems^[21]

Gap Identification

Current Research Gaps

Lack of Integrated Systems: Existing research treats model monitoring and prompt optimization as separate problems, missing opportunities for synergistic approaches.^{[1] [11]}

Limited Real-Time Optimization: Most prompt optimization techniques work offline, failing to leverage real-time telemetry for continuous improvement.^{[9] [10]}

Insufficient Telemetry Integration: Current monitoring solutions don't effectively integrate system telemetry with model-specific metrics for holistic optimization.^{[6] [4]}

Scalability Limitations: Existing approaches don't address the scalability challenges of monitoring and optimizing hundreds of models simultaneously.^{[3] [2]}

Problem Statement Identification

The gaps identified lead to a clear problem statement: **"How can we develop an integrated telemetry-aware system that automatically monitors ML model performance and optimizes prompts in real-time production environments?"**

This problem addresses the need for systems that can :^{[6] [1]}

- Collect comprehensive telemetry from ML applications
- Analyze performance patterns using AI-driven techniques
- Automatically optimize prompts based on telemetry insights
- Scale to monitor multiple models simultaneously
- Provide actionable insights for continuous improvement

The research opportunity lies in bridging the gap between traditional system monitoring and AI-specific optimization needs, creating intelligent systems that can maintain and improve model performance autonomously.^{[1] [11]}

1. <https://arxiv.org/html/2506.11019v1>
2. <https://dagshub.com/glossary/mlops-monitoring/>
3. <https://www.iguazio.com/glossary/drift-monitoring/>
4. <https://blog.kloudmate.com/beyond-logs-unified-observability-with-opentelemetry-in-2024-178de9f1e0f9>
5. <https://dev.to/nareshnishad/day-52-monitoring-llm-performance-in-production-2d7b>
6. <https://galileo.ai/blog/effective-llm-monitoring>
7. <https://www.confident-ai.com/blog/what-is-llm-observability-the-ultimate-llm-monitoring-guide>
8. <https://www.evidentlyai.com/ml-in-production/model-monitoring>
9. <https://www.ibm.com/think/topics/prompt-optimization>
10. <https://arxiv.org/abs/2410.14826>
11. <https://www.arxiv.org/abs/2509.14294>
12. <https://rsdjournal.org/rsd/article/download/42042/34097/446498>
13. <https://www.evidentlyai.com/ml-in-production/data-drift>
14. <https://www.sciencedirect.com/science/article/pii/S2666764922000303>
15. <https://dl.acm.org/doi/10.1145/3534678.3542617>
16. <https://dev.to/craftedwithintent/enhancing-observability-in-machine-learning-with-opentelemetry-insightfulai-update-55p6>
17. <https://onlinescientificresearch.com/articles/the-aidriven-future-of-realtime-telemetry-analytics-in-computer-networks.pdf>
18. <https://opentelemetry.io/blog/2024/llm-observability/>
19. <https://aclanthology.org/2024.emnlp-main.226/>
20. <https://arxiv.org/abs/2405.17346>
21. <https://arxiv.org/abs/2507.14241>
22. <https://www.datadoghq.com/blog/ml-model-monitoring-in-production-best-practices/>
23. <https://blogs.oracle.com/ai-and-datascience/post/training-a-machine-learning-model-on-time-series-telemetry-to-optimize-manufacturing>
24. <https://newrelic.com/blog/nerdlog/ml-model-performance-monitoring>
25. <https://openreview.net/forum?id=Qhc8xDRZuH>
26. <https://orq.ai/blog/prompt-optimization>
27. https://www.reddit.com/r/mlops/comments/15z3bfo/model_performance_in_production/
28. <https://www.ijert.org/prediction-of-telemetry-data-using-machine-learning-techniques>
29. <https://docs.newrelic.com/docs/mlops/get-started/intro-mlops/>
30. <https://arxiv.org/html/2408.02057v1>
31. <https://www.promptingguide.ai/guides/optimizing-prompts>
32. <https://cloud.google.com/architecture/mlops-continuous-delivery-and-automation-pipelines-in-machine-learning>
33. <https://www.observe.ai/post/what-is-telemetry>
34. <https://cameronrwolfe.substack.com/p/automatic-prompt-optimization>
35. <https://mlops-guide.github.io/MLOps/Monitoring/>

36. <https://arxiv.org/abs/2309.03409>
37. <https://www.ibm.com/think/topics/model-drift>
38. <https://learn.microsoft.com/en-us/azure/machine-learning/how-to-monitor-datasets?view=azureml-api-1>
39. <https://www.acceldata.io/blog/data-drift>
40. https://newrelic.com/sites/default/files/2024-10/new-relic-2024-observability-forecast-report_0.pdf
41. https://www.splunk.com/en_us/blog/learn/llm-monitoring.html
42. <https://docs.aws.amazon.com/sagemaker/latest/dg/model-monitor.html>
43. <https://sapphireventures.com/blog/observability-in-2024-understanding-the-state-of-play-and-future-trends/>
44. <https://langfuse.com/faq/all/llm-observability>
45. <https://logz.io/blog/observability-engineering/>
46. https://www.reddit.com/r/LLMDevs/comments/1feb51m/how_do_you_monitor_your_llm_models_in_prod/
47. <https://www.sciencedirect.com/science/article/pii/S0148296319304564>
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